🛡️ SOC Internship – Day 14 Notes

🔍 Threat Hunting Basics (Advanced)

📌 1. What is Threat Hunting?

- Threat hunting is a **proactive cybersecurity approach**
- SOC analysts **actively search for hidden threats**
- Focus is on detecting attacks that **do NOT trigger alerts**

⚖️ 2. Monitoring vs Threat Hunting

Monitoring	Threat Hunting
Reactive approach	Proactive approach
Waits for alerts	Searches for threats manually
Uses SIEM alerts	Uses hypothesis & investigation

🧠 3. Hypothesis-Based Hunting

A structured way to hunt threats:

1. **Create Hypothesis**
    a. Example: "An attacker is using brute force login attempts"
2. **Search Logs**
    a. Check authentication logs, system logs, etc.
3. **Validate Findings**
    a. Confirm whether suspicious activity is real or false positive

💡 4. Example Scenario

Hypothesis:

- Attacker is targeting admin accounts

Observed Logs:

- Multiple login attempts from different IPs

- Login attempts at unusual times (e.g., midnight)

Finding:

- Possible **credential attack / brute force attack**

🛠️ 5. Common Threat Hunting Techniques

- 🔎 Detect unusual login times
- 🔁 Identify repeated failed login attempts
- 🌐 Monitor unknown or suspicious IP addresses
- 👤 Analyze abnormal user behavior

👮 6. SOC Roles in Threat Hunting

- **L2 Analyst**
  - Assists in investigation
  - Analyzes alerts and logs
- **L3 Analyst**
  - Leads threat hunting activities
  - Creates hypotheses and deep analysis

📚 Assignment (Advanced)

1. Create a **threat hunting hypothesis**
2. Analyze logs to find patterns
3. Document and write your **findings**